

Equinox

Brisbane Boys' College
Standard Soccer

Interesting / Innovative Feature

The most interesting and innovative feature of our robot is its movement function. The function uses trigonometry based on the balls angle to accumulate four motor speeds. Combined with an offset that allows for us to effectively orbit the ball.

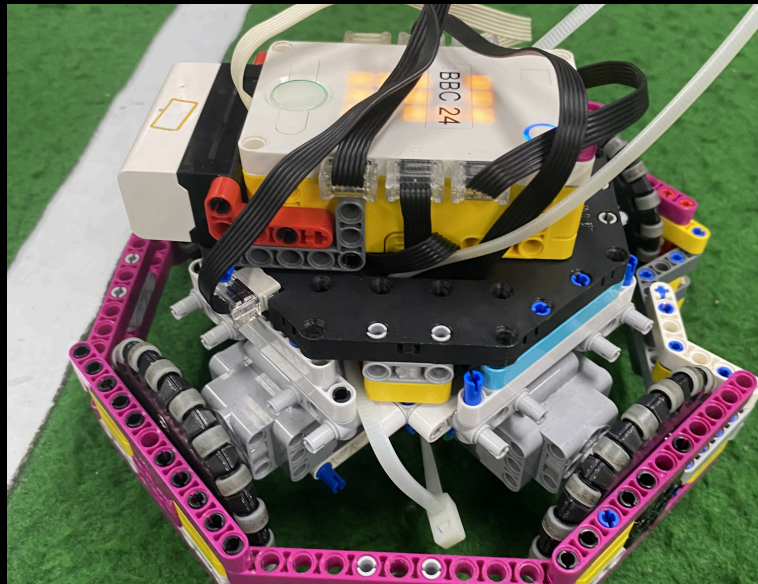


Image of our defensive robot

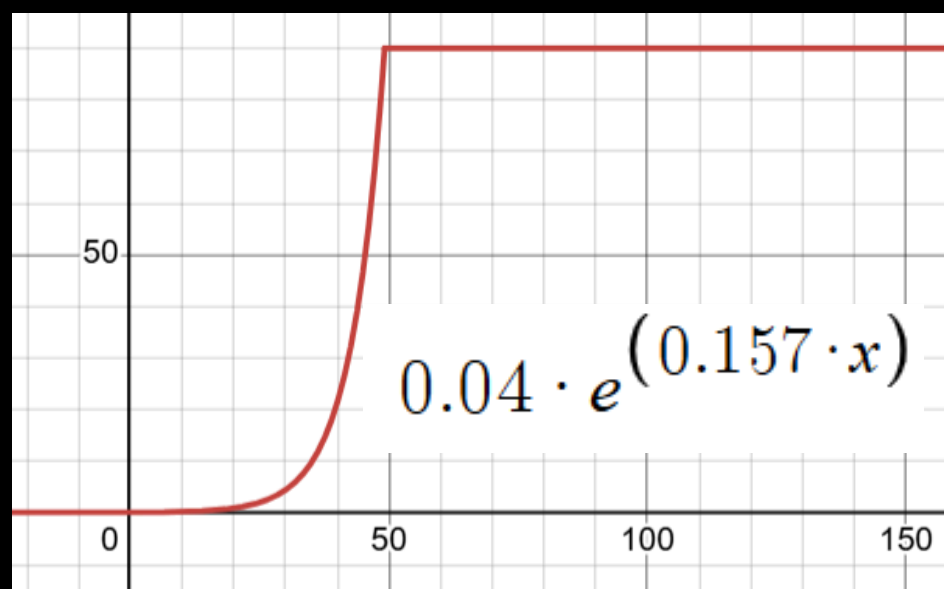


Image of our orbit additional equation

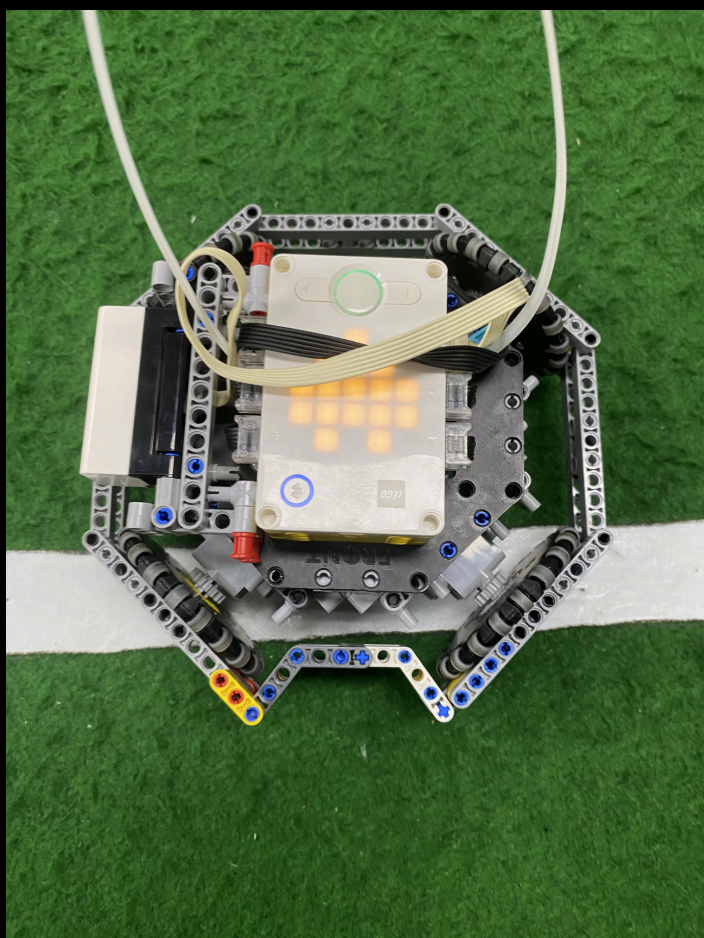


Image of our offensive robot

The Challenge

One of the biggest challenges we faced was a problem with our defensive charging. The problem was that the requirements for our robot was too broad and the readings from the sensors were inaccurate. The robot was charging randomly because of the requirements. We fixed this by being more specific with the requirements as well as lowering the speed of the charge at greater distances. The problem is yet to be fully solved but is at a point where it doesn't majorly jeopardise us.

Lessons Learned

The lesson we learned at robocup regionals was regarding our compass correction segment of our code. We found out that the adjustment multiplier was not strong enough as well as how some motors require the adjustment to be subtracted instead of added. Since then we have fixed that problem!

Members:

- Sam - Coder
- Amaru - Builder & Coder